St. Louis Pipe & Supply
CONFIDENTIAL

vallourec

VALLOUREC STAR, LP
Seamless Tubular Products

Certified Test Report
11/28/2023

Order Line #: O-221125736-1
Desc: SEAMLESS QUENCH AND TEMPERED
S LN (Heat): N90101
Grade: API 5L X52Q 46th Edition May 1, 2019
W/O No: DE7158
OD (in): 10.750 **Wall (in):** 0.500 **Lbs/Ft:** 54.79 **Length:** DRL
Lot: N638910
End Finish: PEB
Comment: Melted and Manufactured in the USA
Vallourec Star s QA program meets the req. of DIN EN 10204-2004 Type 3.1 Satifies NACE MR0175 PAR3.2/ISO 15156-2: 2020 & ANSI/NACE MR0103/ISO 17945:2015
X52Q/LT ASTM A333-18 G1&G6 ASME SA333-2021 G1&G6 ASTM A106B/C-19a ASMESA106B/C-2021 ASTM A53B-22 ASME SA53B-2021 MEETS PSL2

Mechanical Properties

		Tensile Size Specimen Cross Section			Strength (ksi)			Elongation	
Orientation	Type	Width (in)	Thick (in)	Area (sqin)	Yield	Tensile	Y/T Ratio	Gage Length	%Elong
LONGITUDINAL	STRIP	1.493	0.507	0.7597	65.0	77.8	0.84	2	44.0
LONGITUDINAL	STRIP	1.495	0.483	0.7247	66.5	78.7	0.85	2	43.0
LONGITUDINAL	STRIP	1.494	0.491	0.7362	65.7	77.8	0.84	2	44.0
Calculated Max Internal Yield (psi): 5294									

Supplemental Requirement

Austenitizing Temp	1700 °F
Tempering Temp	1315 °F
Flattening	PASSED
Hardness	SEE ATTACHED
Collapse Min	N/A
Collapse Actual	N/A
Hydrostatic Pressure	3,000 5 SEC HOLD
Other	N/A

Chemical Analysis

Chem Spec: 69Q

NDE SEE ATTACHED

WT%	C	Mn	P	S	Si	Cu	Ni	Cr	Mo	Sn	Nb	V	Al	Ca	B	Ti	N	C.E.
HEAT	0.11	1.09	0.012	0.004	0.23	0.15	0.06	0.13	0.02	0.010	0.000	0.044	0.026	0.0014	0.0005	0.003	0.0096	0.34
PRODUCT 1	0.11	1.05	0.011	0.004	0.23	0.14	0.06	0.13	0.02	0.009	0.000	0.044	0.027	0.0016	0.0004	0.003	0.0085	0.34
PRODUCT 2	0.11	1.05	0.011	0.004	0.23	0.15	0.06	0.13	0.02	0.009	0.000	0.044	0.027	0.0016	0.0004	0.003	0.0086	0.33

Charpy Impact Testing

	Notch	Direction	Size	Temp °F	Position	Ft Lbf 1	Ft Lbf 2	Ft Lbf 3	Ft Lbf Avg	Shear 1	Shear 2	Shear 3	Shear Avg
Min Target	–	TRANS	FULL	32	–	15	15	15	20				
Actual	V	TRANS	FULL	32	TRAIL	320	325	322	322	100	100	100	100
Min Target	–	TRANS	FULL	-50	–	10	10	10	13				
Actual	V	TRANS	FULL	-50	TRAIL	241	266	253	253	100	100	100	100
Actual	V			-50	TRAIL	241	266	253	253	100	100	100	100

DRAFT

NDT Inspection: WMSEA

This material has been produced and tested in accordance with the requirements of applicable specifications unless otherwise listed below. We hereby certify that the above test results are representative of those contained in the records of the company. Any modification to this certification as provided by Vallourec Star, without the expressed written consent of Vallourec Star negates the validity of the test report. Vallourec Star is not responsible for the inability of the material to meet specific applications.

Signed:

Name: PETER YANG
Date Approved: 11/28/2023

Production Bend Report

Tulsa Tube Bending

Customer:	DNOW	**Report No.:**	105093-PBR-1
Customer PO No.:	662682056	**Code of Fabrication:**	
TTB Job No.:	25H105093	ASME B16.49-2023	

Mother Pipe & Bend Information

OD (in): 10.750 Grade: API-5L-X52 PSL 2 Radius (in): 50 Bending Machine ID: M12 Heat No.: N90101

Essential Variables

Wall Thickness (inch): 0.500 Radius-to-Diameter Ratio: 5.0 Coil Design: TTB High Spec Coolant Type: Water

Weld Seam Location: n/a Heat Treatment Holding Temp (°F): 1050 Heat Treatment Soak Time (min): 60.0

Note 1: Bending recipe (Bend Speed & Bend Temp parameters) are proprietary to Tulsa Tube Bending Co. and can be made available to the purchaser or the purchaser's third party inspector(s) at the bend facility.

Bend ID No.	Operator Initials	Date	Bending Speed (ipm)	Extrados Bend Temp (°F):	Intrados Bend Temp (°F):	Coolant Temp (°F)	Coolant Flow Rate (GPM)	Induction Frequency (Hz)
Qualification Bend Value:			See Note 1.			72.3	10.0	871
Limits of Variation:			+/- 0.1 ipm	+/- 50°F	+/- 50°F	+/- 25°F	+/- 10%	+/- 20%
105093-1-1	MH	* 4/14/2025	+/- 0.1 ipm	+/- 50°F	+/- 50°F	69.0	10.0	880
105093-1-2	MH	* 4/14/2025	+/- 0.1 ipm	+/- 50°F	+/- 50°F	70.0	10.0	881
105093-1-3	MH	* 4/14/2025	+/- 0.1 ipm	+/- 50°F	+/- 50°F	70.0	10.0	880
105093-1-4	MH	* 4/14/2025	+/- 0.1 ipm	+/- 50°F	+/- 50°F	70.0	10.0	880
105093-1-5	MH	* 4/14/2025	+/- 0.1 ipm	+/- 50°F	+/- 50°F	71.0	10.0	882
105093-1-6	MH	* 4/14/2025	+/- 0.1 ipm	+/- 50°F	+/- 50°F	72.0	10.0	881
105093-1-7	MH	* 4/14/2025	+/- 0.1 ipm	+/- 50°F	+/- 50°F	71.0	10.0	882
105093-1-8	MH	* 4/14/2025	+/- 0.1 ipm	+/- 50°F	+/- 50°F	72.0	10.0	880
105093-2-1	MH	* 4/14/2025	+/- 0.1 ipm	+/- 50°F	+/- 50°F	71.0	10.0	881
105093-2-2	MH	* 4/14/2025	+/- 0.1 ipm	+/- 50°F	+/- 50°F	72.0	10.0	880

* Revised 5/22/25 to reflect actual date of production.

Justin Carter 5/22/25

Customer Information
Your PO 25H105093-1

105093-1-1

105093-1-2

105093-1-3

105093-1-4

105093-1-5

105093-1-6

105093-1-7

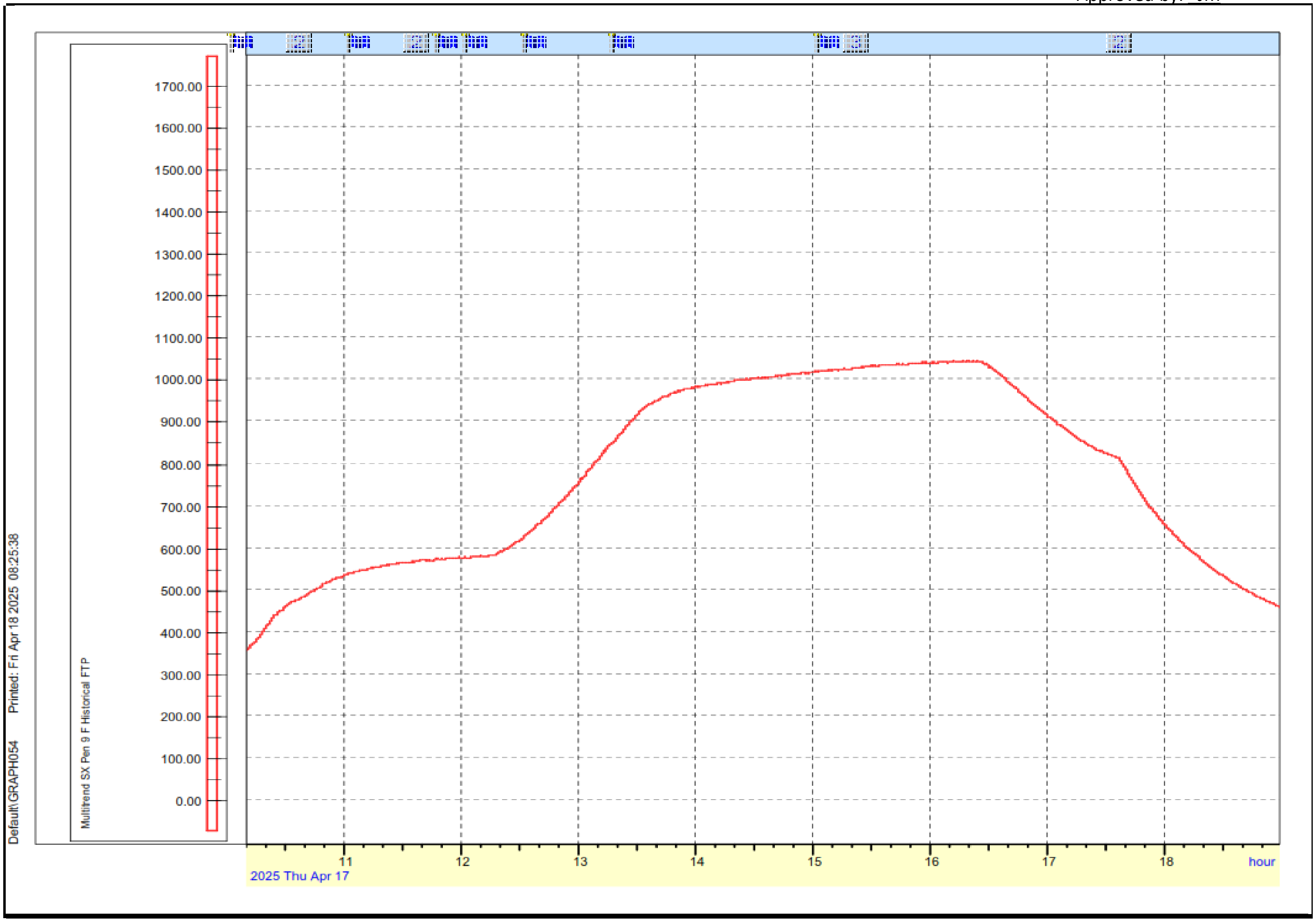
105093-1-8

105093-2-1

105093-2-2

Thermal Specialties
Heat Treating Certification

Thermal Specialties. Information
Date Processed: 04/17/2025
Order ID: 299356 tc load
Furnace#: 27
Calibration Date: 11/19/2024
Thermocouple Type K
Approved by: _JM_



Thermal Specialties certifies that the item has been processed in accordance with customer's requirements.
Processes were obtained through standard approved methods and in accordance with Thermal Specialties quality assurance manual

Production Bend Hardness Report

Tulsa Tube Bending

Customer:	DNOW	**Report No.:**	105093-H-1
Customer PO No.:	662682056	**Code of Fabrication:**	
TTB Job No.:	25H105093	ASME B16.49-2023	

Mother Pipe Information

Pipe Size (in): 10.75 x 0.500

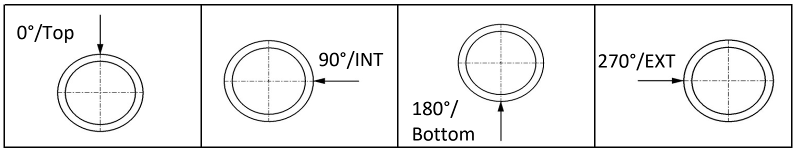
Grade: API-5L-X52 PSL 2

Process Information

Instrument Model No.: GE MIC-10

Serial No.: 1-1718

Key:



T= Top (Weld if Welded Material)

H= Weld Heat Affected Zone

I= Intrados of Bend

E= Extrados of Bend

TAN= Tangent of Bend

STZ= Start Transition Zone

FTZ= Finish Transition Zone

3/4= 3/4 Point of the Bend Arc

Note 1. Weld & Weld HAZ readings are only required on bends made from welded mother pipe.

Note 2. Readings at 3/4 bend arc locations are only required for production bends at least 30° greater than the QB angle.

Hardness Readings

Min Allowed Values: 137 Max Allowed Values: 238

Bend ID No.	TAN T 0°	TAN H 0°	TAN 270°	STZ E 270°	STZ I 90°	MID T 0°	MID H 0°	MID E 270°	MID I 90°	FTZ E 270°	FTZ I 90°	3/4 T 0°	3/4 H 0°	3/4 EXT 270°	3/4 INT 90°
105093-1-1	N/A	N/A	190	180	180	174	N/A	179	176	176	186	N/A	N/A	N/A	N/A
105093-1-2	N/A	N/A	189	179	175	180	N/A	179	169	170	170	N/A	N/A	N/A	N/A
105093-1-3	N/A	N/A	191	170	179	176	N/A	179	176	171	183	N/A	N/A	N/A	N/A
105093-1-4	N/A	N/A	178	169	170	170	N/A	176	168	173	171	N/A	N/A	N/A	N/A
105093-1-5	N/A	N/A	185	173	177	177	N/A	178	175	172	180	N/A	N/A	N/A	N/A
105093-1-6	N/A	N/A	178	180	178	174	N/A	181	181	177	175	N/A	N/A	N/A	N/A
105093-1-7	N/A	N/A	185	179	172	185	N/A	173	180	173	174	N/A	N/A	N/A	N/A
105093-1-8	N/A	N/A	179	179	175	184	N/A	180	178	180	173	N/A	N/A	N/A	N/A
105093-2-1	N/A	N/A	175	175	171	178	N/A	180	175	173	175	N/A	N/A	N/A	N/A
105093-2-2	N/A	N/A	177	180	189	178	N/A	179	182	177	179	N/A	N/A	N/A	N/A

READINGS: ☑ MEET ACCEPTANCE CRITERIA **PERFORMED BY:** JV **DATE:** 4/22/2025

☐ DO NOT MEET ACCEPTANCE CRITERIA

Dimensional-Visual Inspection Report

Tulsa Tube Bending

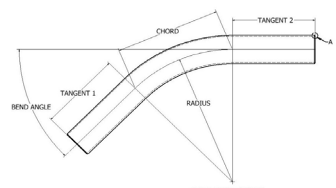
Customer:	DNOW	**Pipe Size (in):**	10.75 x 0.500	**Report No.:**	105093-DV-1-7
Customer PO No.:	662682056	**Grade:**	API-5L-X52 PSL 2	**Bend ID No.:**	105093-1-7
TTB Job No.:	25H105093	**Heat No.:**	N90101	**Code of Fabrication:**	ASME B16.49-2023

Physical Dimensions

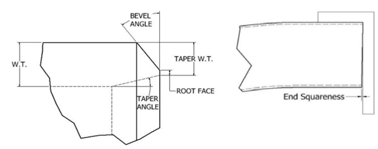
Dimension	**Target**	**Tolerance**	**Actual**
Tangent 1 (in)	24 MIN	MIN	25.88
Tangent 2 (in)	24 MIN	MIN	25.75
Chord (in)	25.9	n/a	25.8
Angle (deg)	30.0	+/- 0.5 °	30.0
Radius, R (in)	50	+/- 1%	49.7
Performed By:	BB	**Date:**	4/24/2025

End Prep Dimensions

Dimension	**Target**	**Tolerance**	**Actual**	
Bevel Angle (deg)	37.5	+/- 2.5°	37.5	END 1
Root Face (in)	0.06	+/- 0.03 in	0.060	END 1
Taper Angle (deg)	n/a	n/a	n/a	END 1
Min ID	9.75	+/- 0.1 in	9.750	END 1
Max ID	9.75	+/- 0.1 in	9.800	END 1
Max End Sq. (in)	0	+/- 0.09 in	0.030	END 1
Bevel Angle (deg)	37.5	+/- 2.5°	37.500	END 2
Root Face (in)	0.06	+/- 0.03 in	0.030	END 2
Taper Angle (deg)	n/a	n/a	n/a	END 2
Min ID	9.75	+/- 0.1 in	9.750	END 2
Max ID	9.75	+/- 0.1 in	9.800	END 2
Max End Sq. (in)	0	+/- 0.09 in	0.030	END 2
Performed By:	BB	**Date:**	4/24/2025	



BEND DIMENSIONS



END PREPARATION DETAILS (A)

Ovality [(O.D. max – O.D. min)/O.D. nominal] x 100

Quadrant	**Limit**	**Min OD**	**Max OD**	**% Ovality**
End 1	1.0%	10.740	10.764	0.2%
Start of Bend	3.0%	10.662	10.749	0.8%
1/4 of Bend Arc	3.0%	10.665	10.716	0.5%
1/2 of Bend Arc	3.0%	10.667	10.720	0.5%
3/4 of Bend Arc	3.0%	10.697	10.710	0.1%
Stop of Bend	3.0%	10.679	10.717	0.4%
End 2	1.0%	10.748	10.753	0.0%
Performed By:	BB		**Date:**	4/24/2025

Wall Thinning of Extrados

Quadrant	**Limit**	**Nominal Wall Thickness**	**Extrados Wall Thickness**	**% Wall Thinning**
Start of Bend	10.0%	0.500	0.466	6.8%
1/4 of Bend Arc	10.0%	0.500	0.451	9.8%
1/2 of Bend Arc	10.0%	0.500	0.452	9.6%
3/4 of Bend Arc	10.0%	0.500	0.452	9.6%
Stop of Bend	10.0%	0.500	0.462	7.6%
Performed By:	BB		**Date:**	4/24/2025

ID Gauge Drift/Pig Inspection

Minimum Gauge Diameter =	97.0%	of Nominal ID, or	9.458 in
Gauge Length Required =	9.75 in		

Gauge passes through the entire bend without power equipment assistance

Performed By:	BB	**Date:**	4/24/2025

Visual Inspection

All accessible surfaces found to be free of visible injurious defects.
Bend is stenciled with correct information.

Performed By:	BB	**Date:**	4/24/2025

READINGS:

- [✓] MEET ACCEPTANCE CRITERIA
- [ ] DO NOT MEET ACCEPTANCE CRITERIA

Globe X-Ray Services, Inc.
8441 S. Union Ave.
Tulsa, OK 74132
918-446-1696

Certification # 186196

Magnetic Particle Examination Record

Customer: Tulsa Tube Bending Date: 4//2025

Job/P.O. No.: 105093 Reference Code/Specification: ASME SEC V

MT Procedure No. GXS IV - C Rev. No. 1/24 Acceptances: ASME SEC VIII Div. I

Magnetizing Equipment: Parker Contour Probe B300 Serial No. 1861 Technique: X Yoke ______ Prods

Current Type: X A/C ______ D/C Magnetic Particle Type: ______ Wet X Dry ______ Fluorescent X Visible

Amperage (Prod Technique Only) __________ Amps/inch of Prod Spacing.

Part/Item Material Type: Carbon Material Thickness: Varies

Lighting Equipment: ______ Fluorescent Light Model # ____________ Serial # ______________
X White Light using Parker Research Y300 or Y400 Probe light
______ Natural White Light
______ Other: ______________________________

Light Intensity on surface to be examined: 100 Min foot candles N/A µW/cm²

Comments: Test The Entire Extrados (Back Half) of the Bend, from N.A. to N.A., Including the Weld Seam

Examination Results

Qty:	Part Identification	Accept	Reject	Indication Description
10	10.75” O.D. X 0.5” Wall, 50” CLR, API-5L-X52 PSL 2 SMLS Pipe HT# N90101 Bend ID # 105093-1-1, 105093-1-2, 105093-1-3, 105093-1-4, 105093-1-5, 105093-1-6, 105093-1-7, 105093-1-8, 105093-2-1, 105093-2-2	X		None

Examine: Jaime Vera Qualification Level: II

Tulsa Tube Bending
4192 S. Galveston Ave.
Tulsa, OK 74107
Tel: 918-446-4461
Fax: 918-446-1595
Serving others. Building people. Pursuing excellence.

Certificate of Conformance

Customer:	DNOW	Certificate No.:	20250424
Customer PO No.:	662682056	Issue Date:	4/24/2025
TTB Job No.:	25H105093		
Material Size, Grade:	10.75 inch OD x 0.500 inch w.t., API-5L-X52 PSL 2 SMLS	Quantity of Bends:	10
Material Heat No(s):	N90101		

Bend ID Nos.:	105093-1-1	105093-1-2	105093-1-3	105093-1-4	105093-1-5
	105093-1-6	105093-1-7	105093-1-8	105093-2-1	105093-2-2

We hereby certify that the aforementioned induction bends have been inspected and conform to all of the requirements of the Purchase Order. The induction bends have been manufactured in conformance with:

ASME B16.49-2023

*With the following notes and exceptions:

- Maximum wall thinning shall not exceed 10% from nominal.
- Bends to be beveled per ASME 37.5° (±2.5°) per B16.49/B16.25.

Approved By: Austin Carter Date: 4/24/2025

FBE Coating per customer coating spec Doc ID TPE_26907sp.

*With the following notes and exceptions:

- Section 9.0 – Laboratory testing not included.
- Section 6.1 – Due to the hand applied nature of the application, max DFT may exceed 36 mils in isolated areas.
- Section 6.2.9 – Acid wash is not available.
- Section6.3.2 – Heat indicating Temp Stick may be used but will be applied near the cutback area with use kept to a minimum.

Keith Industries
20701 FM 521 Rosharon, TX 77583
+1-713-991-5670

COATING INSPECTION RECORD

KF-010 FBE 5/27/22 REV2

CUSTOMER: TULSA TUBE BENDING	DATE: 4/29/2025	ITEMS: 5 PCS
CUSTOMER PO: 25H105093	PAINT SPEC: CONSUMERS ENERGY COMPANY	

SURFACE PREPARATION

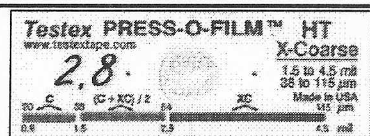
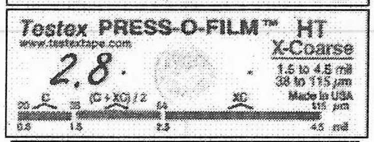
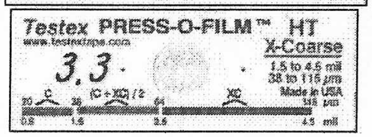
ITEM SALT TEST	N/A☐	RESULT: 0	SSPC SP-1 IF NEEDED		DATE	4/29/2025	4/29/2025
ABRASIVE SALT TEST	N/A☐	RESULT: 1	CLEANLINESS REQUIRED	SSPC SP-10	TIME	10:00AM	12:00PM
PH OF ABRASIVE	N/A☒	RESULT:	CLEANLINESS ACHIEVED	☒YES☐NO	AIR TEMP	81°F	80°F
TYPE OF ABRASIVE	STEEL		PROFILE REQUIRED	2-4 MILS	RELATIVE HUMIDITY	65%	69%
NOZZLE PSI	100PSI		PROFILE ACHIEVED	☒YES☐NO	STEEL	85°F	87°F
BLOTTER TEST	☒PASS		SSPC-VIS 1 USED	☒YES ☐NO	DEW POINT	68°F	69°F
DUST FREE SURFACES	☒YES ☐NO		VISUAL INSPECTION	☒YES ☐NO	STAGE OF OPERATION	PRE-BLAST	PRE-COATING

COATING APPLICATION

COAT 1ST	PRODUCT NAME: VALSPAR 2000	BATCH: CV3444JP	TARGET MILS: 16	TARGET PRE COAT TEMP: 450 DEG F

FINAL TESTING

HVST @ 2000 VOLTS PASS ☒	ADHESION KNIFE TEST: PASS ☒	DFT TEST: PASS ☒	ALL REQUIREMENTS MET: PASS ☒

KI#	DESCRIPTION	MIL AVERAGE	PRESS O FILMS	KI#	DESCRIPTION	MIL AVERAGE
1-3	10.75" X .500"WT API5L X52 SMLS 30 DEG BEND	28				
1-6	10.75" X .500"WT API5L X52 SMLS 30 DEG BEND	24				
1-7	10.75" X .500"WT API5L X52 SMLS 30 DEG BEND	26				
2-1	10.75" X .500"WT API5L X52 SMLS 45 DEG BEND	25				
2-2	10.75" X .500"WT API5L X52 SMLS 45 DEG BEND	23				

VERIFIED BY KEITH: John Keith

Keith INDUSTRIES
20701 FM 521 Rosharon , TX 77583
+1-713-991-5670

Coating Inspection Record

KF-010 FBE 5/27/22 REV2

Customer: Tulsa Tube Bending	Date: 4/30/2025	Items: 5 PCS
Customer PO: 25H105093	Paint Spec: Consumers Energy Company	

Surface Preparation

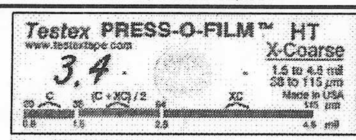
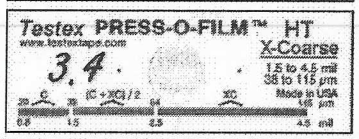
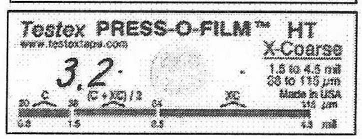
Item Salt Test	N/A☐	Result: 0	SSPC SP-1 IF NEEDED		Date	4/30/2025	4/30/2025
Abrasive Salt Test	N/A☐	Result: 1	Cleanliness Required	SSPC SP-10	Time	10:00AM	12:00PM
PH of Abrasive	N/A☒	Result:	Cleanliness Achieved	☒Yes☐No	Air Temp	80˚F	82˚F
Type of Abrasive	Steel		Profile Required	2-4 mils	Relative Humidity	71%	67%
Nozzle PSI	100PSI		Profile Achieved	☒Yes☐No	Steel	83˚F	88˚F
Blotter Test	☒Pass		SSPC-VIS 1 Used	☒Yes ☐No	Dew Point	70˚F	70˚F
Dust Free Surfaces	☒Yes ☐No		Visual Inspection	☒Yes ☐No	Stage of Operation	Pre-Blast	Pre-Coating

Coating Application

Coat 1st	Product Name: Valspar 2000	Batch: CV3444JP	Target Mils: 16	Target pre coat temp: 450 deg F

Final Testing

HVST @ 2000 Volts PASS ☒	Adhesion Knife Test: PASS ☒	DFT Test: PASS ☒	All requirements met: PASS ☒

KI#	Description	Mil Average	Press o Films	KI#	Description	Mil Average
1-1	*10.75" X .500"WT API5L X52 SMLS 30 DEG BEND*	26				
1-2	*10.75" X .500"WT API5L X52 SMLS 30 DEG BEND*	23				
1-4	*10.75" X .500"WT API5L X52 SMLS 30 DEG BEND*	24				
1-5	*10.75" X .500"WT API5L X52 SMLS 30 DEG BEND*	20				
1-8	*10.75" X .500"WT API5L X52 SMLS 30 DEG BEND*	19				

Verified by Keith: John Keith